

4	(a)		kinetic energy was converted to <u>gravitational</u> potential energy and work done against frictional force.	B1
	(b)		$\frac{1}{2}mv^2 = mg\Delta h + \text{work done against friction}$	M1
			$\Delta h = 150 \sin 0.57^\circ = 1.492 \text{ m}$	C1
			$\text{work done against friction} = \frac{10^2}{2}m - 9.81 \times 1.492m = 35.36m$	M1
			$\text{proportion of energy lost due to friction} = \frac{35.36}{10^2/2} = 0.71$	A1

5	(a)		Since the mass is not moving in a straight line, there must be a net	
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